# COMPREHENSIVE NETWORK SIMULATION AND PERFORMANCE ANALYSIS FOR A MULTI-SITE ENTERPRISE ARCHITECTURE

**Technologies Chosen:**

IoT

SD-WAN

Data Security

**Student Name:** _____

**Student ID:** _____

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# 1. Introduction

The enterprise network of the modern world is growing increasingly dependant on the sophisticated digital technology in order to enhance scalability, openness of communication, and organisational flexibility. The Internet of Things (IoT), Software-Defined Wide Area Networking (SD-WAN), and improved Data Security frameworks are among them and are essential in facilitating effective and robust global business activity.

## 1.1 Upgrade Engineer Perspective – Benefits and Scalability

The **Internet of Things (IoT)** facilitates large scale interconnection between sensors, endpoints, and smart devices; it allows these devices to interact with each other on the networks independently. IoT enhances operational visibility and transparency through provision of real-time data analytics, predictive maintenance and automated decision systems. As an illustration, the AWS IoT Core enables organisations to scale their millions of devices using scalable protocols like MQTT and HTTPS, which improve communication among the globally distributed locations (Dai, 2024). Within the framework of multi-site architecture of the company, IoT can enhance the control over network equipment, monitoring of environmental conditions in remote offices, and monitoring of the work of backbone links.

**SD-WAN** removes the coupling of physical infrastructure and network control, which increases scalability and performance. It provides clear and optimised communication amongst geographically distributed offices, by intelligent traffic steering, centralised control of policies and dynamic path selection. Cloud-enabled SD-WAN infrastructures, including Cisco Viptela and VMware Velocloud, are offering reliable, high-quality access to such applications as MS Teams, OneDrive, and Zoom. The SD-WAN is especially advantageous to this organisation since it will ease the management of the WAN throughout Cape Town, Singapore, Buenos Aires, and the NOC, as well as enhance QoS of cloud-hosted workloads (Gupta et al., 2022).

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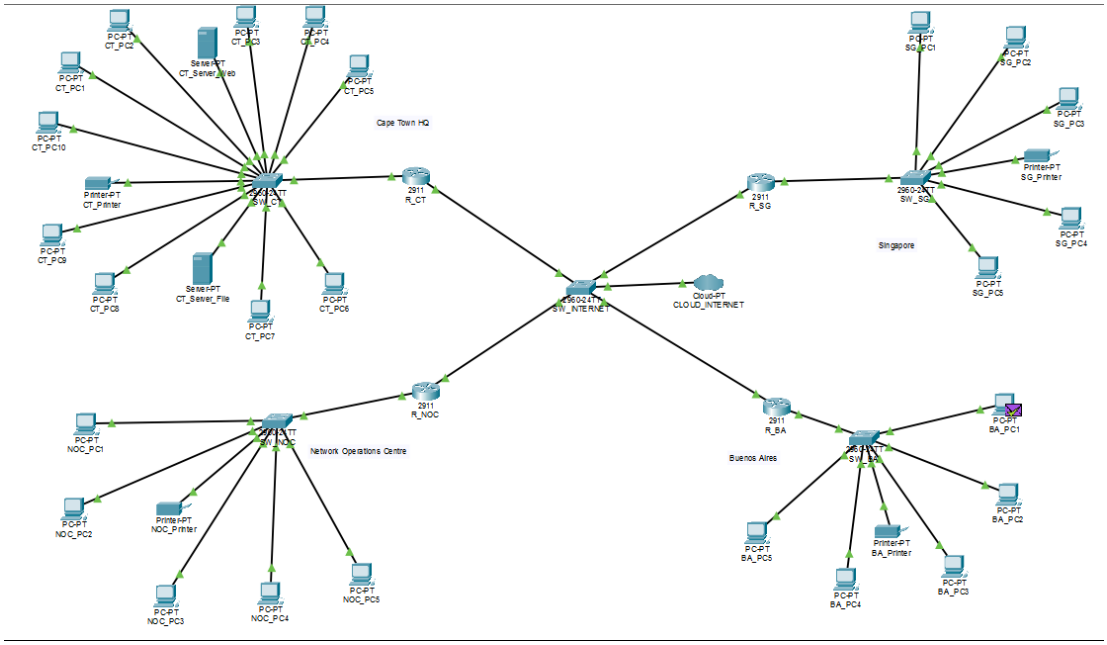


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R_CT

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_CT
R_CT(config)#interface GigabitEthernet0/0
R_CT(config-if)# description LAN_to_SW_CT
R_CT(config-if)# ip address 10.1.0.1 255.255.255.0
R_CT(config-if)# no shutdown

R_CT(config-if)#exit
R_CT(config)#interface GigabitEthernet0/1
R_CT(config-if)# description WAN_to_INTERNET_SWITCH
R_CT(config-if)# ip address 10.255.0.11 255.255.255.0
R_CT(config-if)# no shutdown

R_CT(config-if)#exit
R_CT(config)#end
R_CT#write memory
Building configuration...
[OK]
```

[redacted]

CT_PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.0.1

Pinging 10.1.0.1 with 32 bytes of data:

Reply from 10.1.0.1: bytes=32 time<1ms TTL=255
Reply from 10.1.0.1: bytes=32 time<1ms TTL=255
Reply from 10.1.0.1: bytes=32 time=1ms TTL=255
Reply from 10.1.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

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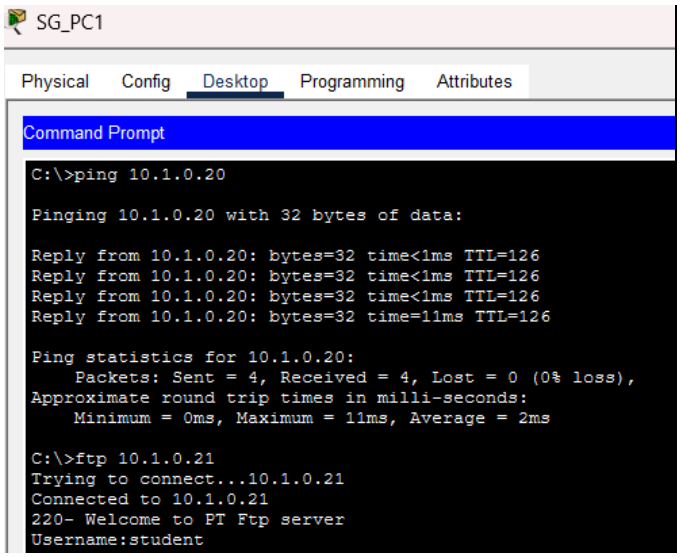


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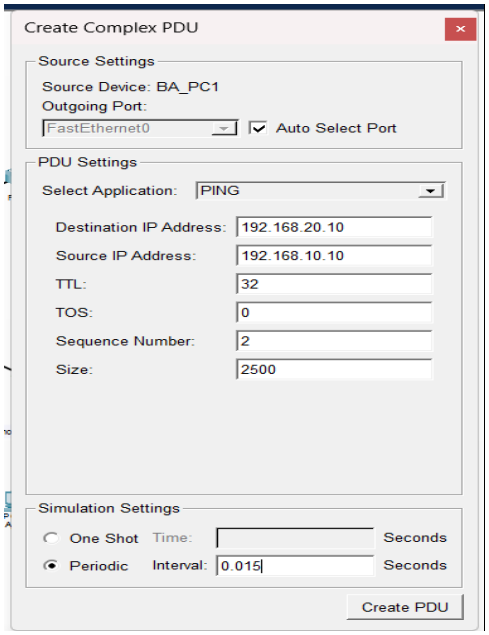


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```
R_CT>enable
R_CT#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.0.1        YES manual up                    up
GigabitEthernet0/1     10.255.0.11     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R_CT#
```

```
R_SG>enable
R_SG#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.1.1      YES manual up                    up
GigabitEthernet0/1     10.255.0.12     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R_SG#
```

```
R_BA>enable
R_BA#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     10.255.0.13     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R_BA#
```

```
R_NOC>enable
R_NOC#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.20.1    YES manual up                    up
GigabitEthernet0/1     10.255.0.14     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R_NOC#
```

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Simulation Panel

Event List

| Vis. | Time(sec) | Last Device | At Device | Type |
|---|---|---|---|---|
| | 0.006 | BA_PC1 | SW_BA | ICMP |
| | 0.007 | SW_BA | R_BA | ICMP |
| | 0.015 | -- | BA_PC1 | ICMP |
| | 0.016 | BA_PC1 | SW_BA | ICMP |
| | 0.017 | SW_BA | R_BA | ICMP |
| | 0.018 | R_BA | SW_INTERNET | ICMP |
| | 0.019 | SW_INTERNET | R_NOC | ICMP |
| | 0.020 | R_NOC | SW_NOC | ICMP |
| | 0.021 | SW_NOC | NOC_PC1 | ICMP |
| | 0.021 | SW_NOC | NOC_PC2 | ICMP |
| | 0.021 | SW_NOC | NOC_PC3 | ICMP |
| | 0.021 | SW_NOC | NOC_PC4 | ICMP |
| | 0.021 | SW_NOC | NOC_PC5 | ICMP |
| | 0.021 | SW_NOC | NOC_Printer | ICMP |
| | 0.022 | NOC_PC1 | SW_NOC | ICMP |
| | 0.023 | SW_NOC | R_NOC | ICMP |
| | 0.024 | R_NOC | SW_INTERNET | ICMP |
| | 0.025 | SW_INTERNET | R_BA | ICMP |
| | 0.026 | R_BA | SW_BA | ICMP |
| | 0.027 | SW_BA | BA_PC1 | ICMP |

Reset Simulation ☑ Constant Delay

Captured to: 0.027 s

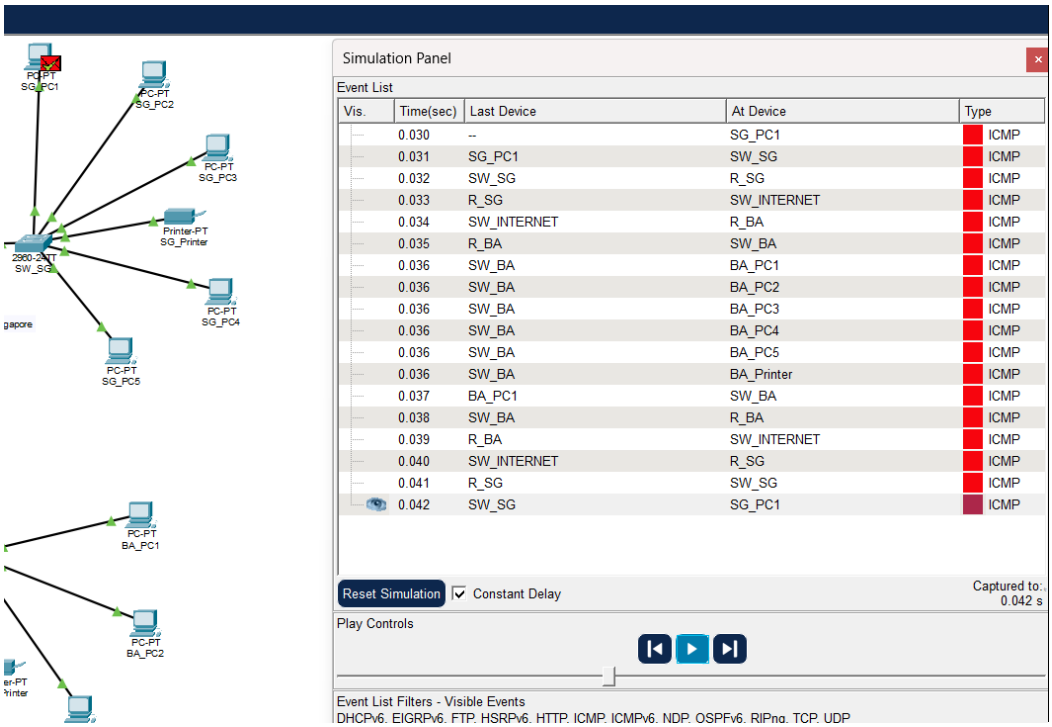
[redacted]

```
ftp>get asa842-k8.bin

Reading file asa842-k8.bin from 10.1.0.21:
File transfer in progress...

[Transfer complete - 5571584 bytes]

5571584 bytes copied in 9.675 secs (131950 bytes/sec)
```

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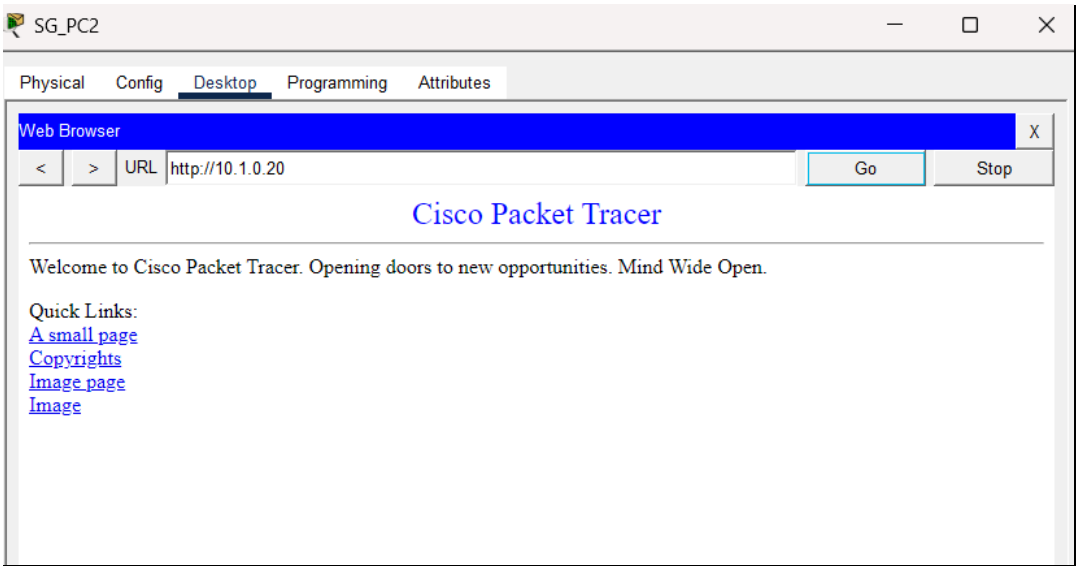


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